

CSE 4327 VLSI Design – Lab Final Viva Cheat Sheet

Know these first (fixes and gotchas)

- **Mux 2:1** finishes at **60 ns** (6 vectors), not 80. **Mux 8:1**: s=010 selects i2 (console: inputs 00000100, out=1). Both fixed in your corrected reports.
- **Counter** wraps 1111 → 0000 at **165 ns**; your waveform only shows 0–80 ns. Re-run to 180 ns before the viva and be ready to show the wrap.
- **JK** first toggle takes effect at the **55 ns** edge (inputs change at 50).
- Your + adders are **behavioral**, not ripple-carry. Test benches use \$monitor (not self-checking) except Exp 7.
- Exp 7: confirm slack / cell count / power from your own rc output (the report's table values are marked “~”). Its SDC was carried over from Exp 6 (names rst/count). NOT report says DC sweep 0–2.0 V but plot ends ≈1.8 V.

Q1 & Q2 – Virtuoso: NOT, NAND, AND

- **Setup**: gpd090, VDD=1.8 V, L=100 nm, Wp=720 nm, Wn=240 nm (3:1). vpulse 0 → 1.8 V, 20 ns / 10 ns width; NAND/AND input B 40 / 20 ns (all 4 combos); rise/fall 50 ps; transient 200 ns (NOT: 100 ns + DC sweep).
- **Inverter**: Vin=0 → PMOS on, Vout=VDD; Vin=1 → NMOS on, Vout=0. PMOS wider because hole mobility is lower → balanced rise/fall, VM=VDD/2 (~0.9 V).
- **NAND**: 2 PMOS **parallel** (pull-up), 2 NMOS **series** (pull-down); output low only when A=B=1. NMOS stack is weaker → slower fall, VM shifts up.
- **AND = NAND (PM0, PM1, NM0, NM1) + inverter (PM2, NM2)**; point to the internal node between stages. One CMOS stage always inverts. 6 transistors.
- **VTC**: DC sweep of Vin, plot Vout; VM is where Vin=Vout. **Flow**: library + tech → schematic → symbol → test bench → ADE L (Spectre, model lib, tran/DC).
- **Swap PMOS/NMOS**: becomes non-inverting with degraded levels (high ≈ VDD–Vtn, low ≈ |Vtp|).

Q3 – 4-to-1 MUX (Verilog)

- **Yours = dataflow**, one assign with nested ?:; say “dataflow”, not structural. Truth table: out = i[s].
assign out = s[1] ? (s[0]?i[3]:i[2]) : (s[0]?i[1]:i[0]);
- **Three 2:1 muxes**: two first-level (s[0]: i0/i1, i2/i3) + one second-level (s[1]). n:1 needs n–1 (8:1 → 7). Live edit:
mux21 m0(.i0(i[0]),.i1(i[1]),.s(s[0]),.out(w0));
mux21 m1(.i0(i[2]),.i1(i[3]),.s(s[0]),.out(w1));
mux21 m2(.i0(w0),.i1(w1),.s(s[1]),.out(out));
- **#10** = 10 time units = 10 ns (timescale 1ns/1ps). Doubling it doubles each hold and the run (2:1 60 → 120, 4:1 80 → 160, 8:1 160 → 320 ns). Combinational: out changes in the same time step as inputs. Console times are in **ps**.

Q4(a) – 4-bit Full Adder

- **Half adder**: sum=a^b, carry=a&b. **Full adder = 2 HA + OR**: sum=a^b^cin, cout=ab+cin(a^b). Yours: (a&b)|(b&cin)|(cin&a), equivalent.
- **You wrote**: 1-bit dataflow; 4–64-bit behavioral {carry,sum}=a+b+cin; subtractor {bout,diff}=a-b-bin (0–15 → 0001, borrow 1); multiplier 8-bit prod=a*b (9×9=81). 15+15+1=31 → sum 1111, carry 1.
half_adder h1(a,b,s1,c1); half_adder h2(s1,cin,sum,c2); or(carry,c1,c2);

Q4(b) – D FF, JK FF, 4-bit counter

- **Latch** = level-sensitive; **FF** = edge-triggered (posedge clk). Sensitivity posedge clk or negedge rst: capture edge + **asynchronous** reset (drop negedge rst → synchronous).
- **<= vs :=**: non-blocking evaluates all RHS first, updates together → no race between flops; use it in clocked blocks.
- **NCLaunch**: cds.lib + worklib → ncvtlog (compile/syntax) → ncelab (hierarchy, snapshot, test bench top) → ncsim + SimVision (run/waveform).
- Clock always #5 clk=~clk → 10 ns period, rising edges at 5, 15, 25... rst released at 10 ns.

- **D FF** (d=1@20, 0@30, 1@40, ends 50): edges 15/25/35/45 → q = 0, 1, 0, 1.
- **JK** (00@10, 10@20, 00@30, 01@40, 11@50, 11@60, ends 70): 15 hold q=0 · 25 set q=1 · 35 hold q=1 · 45 reset q=0 · 55 toggle q=1 · 65 toggle q=0. qbar=~q.
- **Counter** (rst low 0–10): count=n at edge 5+10n (15 → 1, 75 → 7), 1111 at 155, wraps to 0000 at **165**, ends 180 ns.

Exp 6 & 7 – Synthesis (RTL Compiler)

- **Flow**: simulate RTL → SDC → rc -f rc_script.tcl (read_hdl -v2001, elaborate, read_sdc, synthesize -to_mapped -effort medium, report timing/power, write_hdl/write_sdc) → gui_show. -v2001 = ANSI port style.
- **SDC**: clk 10 ns {0 5}; transition 0.1; uncertainty 0.1; input delay rst 1.0; output delay count 1.0. Library gpd045 slow.lib.
- **Counter slack**: 330 ps clk → Q + 1000 ps out delay = 1330 arrival; capture 10000–100 = 9900 → **slack 8570 ps**. Power 3.021 leak + 8367.397 dyn = **8370.418 nW**, 7 cells, SDFFRX4. Simulated counter_4bit; synthesized copy is cnt4bit.
- **Pipelined adder**: 8 stages × 4-bit = 32-bit; latency 9 edges (~90 ns); trades flops/area/power/latency for a short per-cycle path. GL0-12 = sum_pipe flops that are always 0 (stage k has only 4k live bits); GL0-32 = 24 unused flops deleted. Your 80 ns run shows errors=7 MISMATCHes (expected = x while the 9-deep reference chain fills) and no PASS; a run past ~90 ns is needed for PASS.